The empirical recurrence for OEIS A303420: recovering a conjecture that is not written down

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Abstract

OEIS A303420 records "Empirical recurrence of order 58", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 119$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 58, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 58 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A303420 is "Number of nX7 0..1 arrays with every element equal to 0, 1, 2, 3, 4 or 5 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$64, 8192, 972168, 116517744, 13963574592, 1673186560760, 200493397340704, 24024569786901184, \dots$$

The entry states:

> Empirical recurrence of order 58 (see link above)

As of the "Last modified" line on the live entry (Apr 23 2018 10:54:24, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 16384$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 119$ of the 16384, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 119$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 238$ exact terms of the model returns order 58 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{58} c_i \, a(n-i),$$

c_1	90	c_{16}	13601208374934563	c_{31}	8214884069540352269493 2	c_{46}	−39636978
c_2	2841	c_{17}	−40909335369282248	c_{32}	18973842270853085036195 2	c_{47}	−43648956
c_3	81048	c_{18}	−391070922187888312	c_{33}	942616265582085462581 76	c_{48}	−30046191
c_4	784236	c_{19}	−351594951026161556	c_{34}	−494781979796909339577200	c_{49}	10222656
c_5	4592689	c_{20}	3433574008770476454	c_{35}	−851244656052668580135232	c_{50}	13535505
c_6	−102304943	c_{21}	15113046819717488357	c_{36}	−313227632785961624548608	c_{51}	85198759
c_7	−1241644506	c_{22}	−50282297202232863 76	c_{37}	183848778409555307988339 2	c_{52}	−5717206
c_8	−8850189463	c_{23}	−127598621454080558268	c_{38}	252167514077149098839705 6	c_{53}	−1841923
c_9	64051223264	c_{24}	−298815671754416820916	c_{39}	949096343427092547769600	c_{54}	−4672207
c_{10}	804797449893	c_{25}	405387526573165614692	c_{40}	−4212769808035545757095936	c_{55}	−164134
c_{11}	2333520596384	c_{26}	2454556003226394279316	c_{41}	−4938509284244477658802176	c_{56}	1423188
c_{12}	−28652930432621	c_{27}	3295739689029191810044	c_{42}	−2365406772474177363296256	c_{57}	−253698
c_{13}	−176763648668737	c_{28}	−8050181670128118947960	c_{43}	5685619500009556998123520	c_{58}	969813
c_{14}	25732331610740	c_{29}	−27368952006859192891296	c_{44}	6170038774650821827624960		
c_{15}	5155337870825714	c_{30}	−21808053703522291873840	c_{45}	3725631438192785222467584		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 58, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $58 + 4$ terms removed returns order 58 as well, so $\deg q_{\min} = 58 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $58 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 58 that this sequence satisfies — and that family turns out to have exactly one member.

References

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